CNN (CONVOLUTIONAL NEURAL NETWORK)

- ANN can't do this because it has high computation power
- Spatial arrangement is missing
- Overfitting

If I ask you what is in the picture?, you don't try to look at all the LEGO pieces at once.

Instead we:

- First look at small areas maybe a window, a tower, or a door.
- Then you put those in your mind.
- Then, you say it's a castle.

① CONVOLUTION LAYER THE DETECTIVE'S MAGNIFYING GLASS

- This layer slides a small filter across the picture
- The filter might highlight edges colors, textures.
- Example: one filter learns edges while other learns shapes

③ MORE CONVOLUTION + POOLING LAYERS BUILDING UNDERSTANDING

- As the network goes deeper, the deeper layers find objects (a cat, a car, a castle)

② POOLING LAYER THE SHORTCUT

- Pooling just keeps the most important information.

④ FULLY CONNECTED LAYER - THE DECISION MAKER

At the end, CNN flattens everything it learned and connects it like a normal ANN.

70% chance it's a cat
20% " " " dog.
10% " " " rat

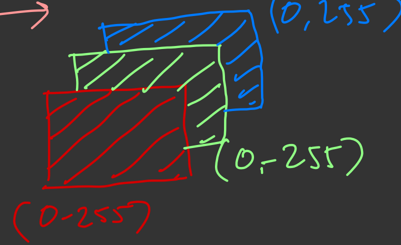
Most prob is the output.

GRAY SCALE IMAGE → Black & white image.

White → 255

Black → 0

RGB Image →



HORIZONTAL FILTER

-1	-1	-1
0	0	0
1	1	1

VERTICAL FILTER

-1	0	1
-1	0	1
-1	0	1

0	0	0	255	255
0	0	0	255	255
0	0	0	255	255
0	0	0	255	255
0	0	0	255	255
0	0	0	255	255

So, we will take this portion and multiply it to the matrix.

$$(0\times -1) + (0\times -1) + (0\times -1)$$

$$(0\times -1) + (0\times -1) + (0\times -1)$$

" + " + "

0	0	0	255	255
0	0	0	255	255
0	0	0	255	255
0	0	0	255	255
0	0	0	255	255
0	0	0	255	255

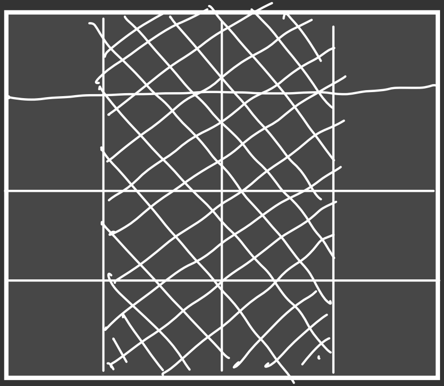
0	0	0	255	255
0	0	0	255	255
0	0	0	255	255
0	0	0	255	255
0	0	0	255	255
0	0	0	255	255

$\times$

-1	0	1
-1	0	1
-1	0	1

0	255	255	0
0	255	255	0
0	255	255	0
0	255	255	0

$\Rightarrow$



VISUALIZATION → deeplizard.com

POOLING

If N=10
The Feature 10 = 3000MB
≈ 3GB

To avoid this we have pooling.

① MAX POOLING

- Dominant features are captured.
- Size issue

② AVG POOLING

- we want smoother, less noisy representation
- less common today, but good for preserving background.

③ MIN POOLING

- highlighting dark features

$$\begin{bmatrix}7&3&5&2\\8&7&1&6\\4&9&3&9\\0&8&4&5\end{bmatrix} \xrightarrow{\text{max}} \begin{bmatrix}8& \\ & \end{bmatrix}$$

$$\begin{bmatrix}7&3&5&2\\8&7&1&6\\4&9&3&9\\0&8&4&5\end{bmatrix} \xrightarrow{\text{max}} \begin{bmatrix}8&6\\ & \end{bmatrix} \Rightarrow \begin{bmatrix}8&6\\9&9\end{bmatrix}$$